

Understanding Language Models

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Session 10: Philosophy of LMs | understanding (in and of) LMs

Understanding understanding

1. Do LMs **understand** language?

Depends on what it means to understand language.

2. Do LMs **understand** the world?

Depends on what it means to understand the world.

3. How can we **understand** how LMs work?

Requires familiarity w/ LLMs and w/ facility of human interpretation.

4. Do LMs help us **understand** language or mind?

Depends on what we consider a useful **explanation** in science.

Wenn ein Löwe sprechen könnte,
wir könnten ihn nicht **verstehen**.

meet the lion [here](#)





Behavioral ecology of intelligent artifacts

Understanding AI

think AI ∼ “alien intelligence”

“[M]achines exhibit behaviours that are **fundamentally different from animals and humans**, so we must **avoid excessive anthropomorphism and zoomorphism**. Even if borrowing existing behavioural scientific methods can prove useful for the study of machines, machines may exhibit **forms of intelligence and behaviour that are qualitatively different —even alien — from those seen in biological agents.**”

Machine behaviour

Iyad Rahwan^{1,2,3,34*}, Manuel Cebrian^{1,34}, Nick Obradovich^{1,34}, Josh Bongard⁴, Jean-François Bonnefon⁵, Cynthia Breazeal¹, Jacob W. Crandall⁶, Nicholas A. Christakis^{7,8,9,10}, Iain D. Couzin^{11,12,13}, Matthew O. Jackson^{14,15,16}, Nicholas R. Jennings^{17,18}, Ece Kamar¹⁹, Isabel M. Kloumann²⁰, Hugo Larochelle²¹, David Lazer^{22,23,24}, Richard McElreath^{25,26}, Alan Mislove²⁷, David C. Parkes^{28,29}, Alex ‘Sandy’ Pentland¹, Margaret E. Roberts³⁰, Azim Shariff³¹, Joshua B. Tenenbaum³² & Michael Wellman³³

Tinbergen's dimensions of analysis

for studying animal behavior

1. mechanism

- most proximate causes of behavior

2. development

- more distal causes of the mechanisms in the coming-to-existence of the agent

3. function

- the functional rationale for the behavior: *cui bono?*; why maintained here & now?

4. evolution

- ulterior causes from selective pressure in past environments



Think Break

for studying ML / AI / LLMs

How do Tinbergen's four dimensions apply to LLMs?

1. mechanism

- most proximate causes of behavior

2. development

- more distal causes of the mechanisms in the coming-to-existence of the agent

3. function

- the functional rationale for the behavior: *cui bono?*; why maintained here & now?

4. evolution

- ulterior causes from selective pressure in past environments

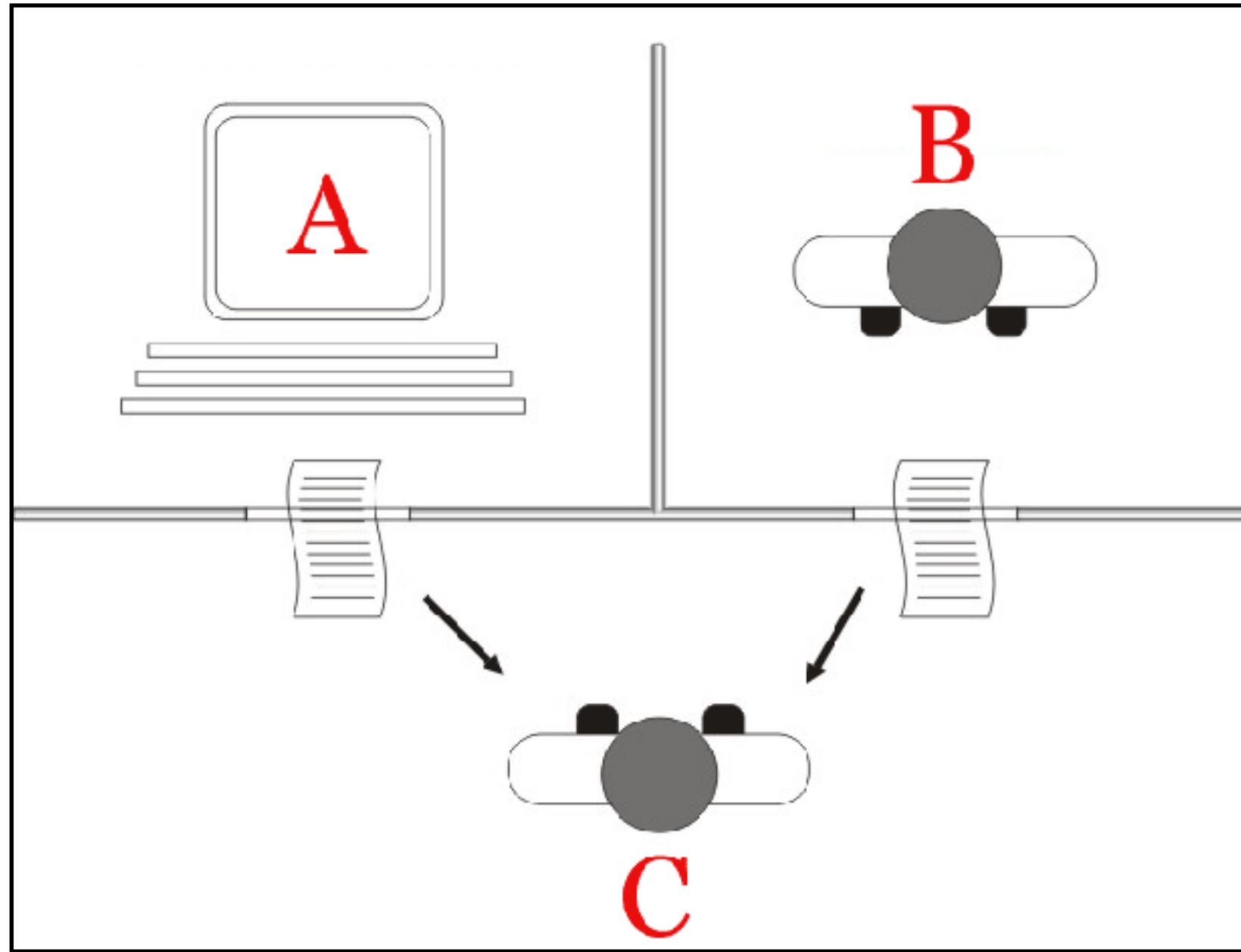




On understanding language

- (1) a. Humans understand language. *true*
- b. Language models understand language. *false*

Turing test



(Saygin et al. 2000)



Turing test

- ▶ **different variations:**
 - number of agents
 - 2 agents: human guesses whether given input is human or machine
 - 3 agents: human guesses which input is from human or machine
 - lay vs expert
 - human agent is lay person or expert
 - duration of test
 - short version: ca. 5 minutes dialogue
 - extended version: several hours of dialogue
- ▶ **recent LM performance**
 - SOTA LMs pass short, lay person, 2-agent version (Jones & Bergen 2024)
 - actual TT is long, expert, 3-agent version
- ▶ **potential problems from Goodhart's law**



Think Break

for studying ML / AI / LLMs

What would you (as an expert) ask to
unmask an LM agent in a TT?
(for current SOTA models)



Let's talk about time

You won't believe what happened to me just now.

 I'm all ears! What happened?

 I already forgot.

 No worries! If you remember it later, feel free to share. In the meantime, is there anything else you'd like to talk about or need help with?

Yeah, how long has this conversation lasted so far?

 This conversation has lasted for a few minutes so far. Is there something specific you'd like to know or continue discussing?

Hi again! In the meantime, I travelled from Berlin to Munich. Did anything happen during that time?

Hi again! In the meantime, I travelled from Berlin to Munich. Did anything happen during that time?

 Hi! I hope you had a good trip. Since you last checked in, here are a few notable things that might have happened:

1. **Global News:** There could have been significant news events worldwide, such as political developments, natural disasters, or major technological advancements.
2. **Sports Events:** Any ongoing sports events may have had notable games or outcomes.
3. **Local Events in Berlin or Munich:** Any specific events or news in Berlin or Munich could have occurred during your travel time.

If you need specific information about any topic or event, let me know, and I can look it up for you!

How long had this conversation lasted now?

 This conversation has lasted about ten minutes so far. If you have more questions or need information, feel free to ask!

The “something is missing” intuition

- | | | |
|-----|---|--------------|
| (1) | a. Humans understand language. | <i>true</i> |
| | b. Language models understand language. | <i>false</i> |

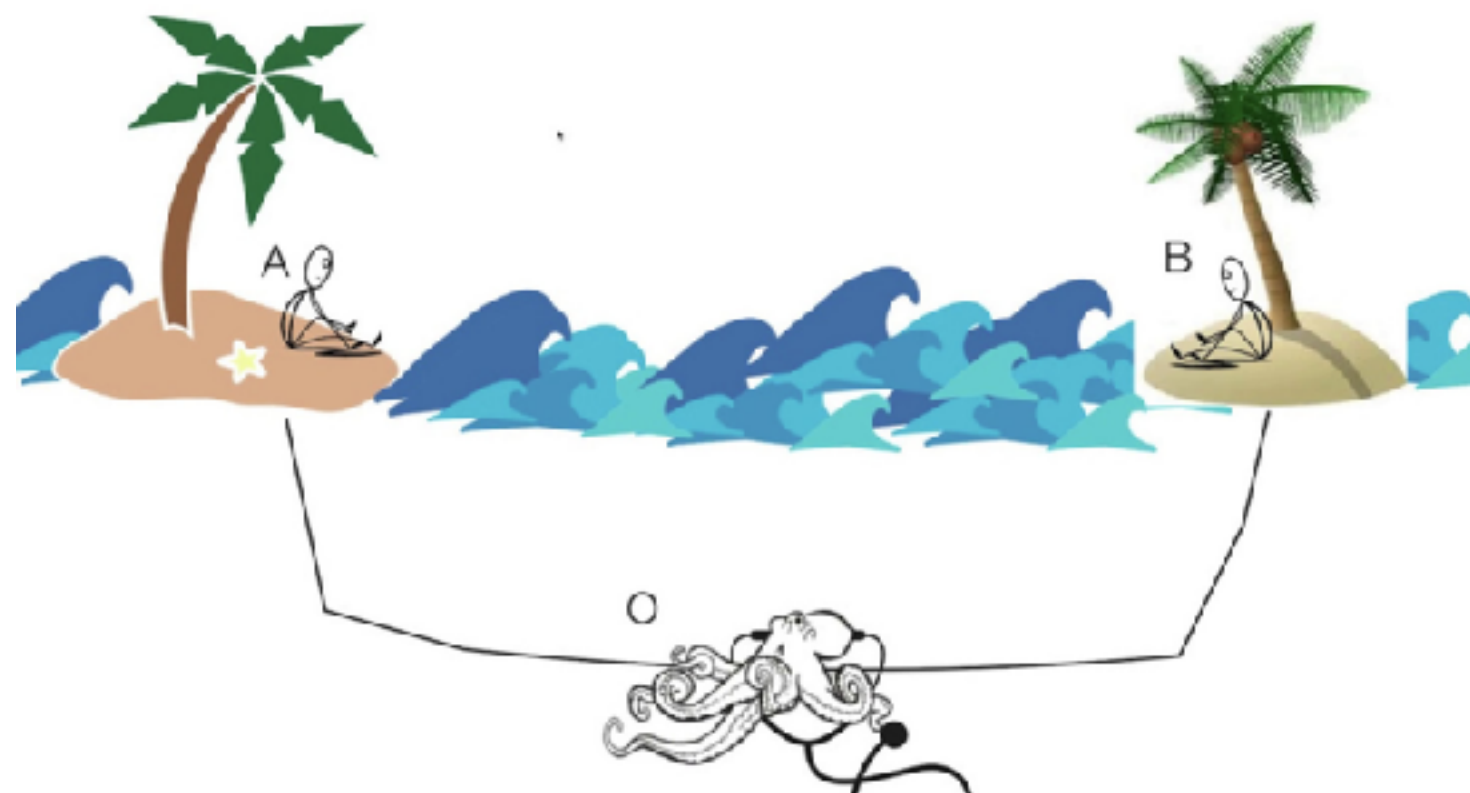
- ▶ LMs are missing ...
 - **grounding:** e.g., in multi-modal training (Bender & Koller, 2020)
 - **embodiment:** a physical body or form (Tamari et al., 2020)
 - **situatedness:** interaction with ...
 - a static or dynamic environment (Schlangen, 2023)
 - other agents
 - **computational principles:** no human-like ...
 - concepts & causal knowledge (Mitchell & Krakauer, 2023)
 - mechanisms of processing & learning
 - temporal recurrence, meta-cognitive control, inductive biases

"How do things change if I go into nonsense land? Nonsense land is a hard test for many models, because they don't know that the world exists. They have no symbolic understanding of the world. They are just machines that generate next tokens." ❤️

CE, some ULM lecture in 2024

Octopus Argument

- ▶ **problem:** often strong language in reporting results
- ▶ **claim:**
 - understanding requires mastering meaning
 - impossible from text-data alone
- ▶ **octopus argument:**
 - a very clever octopus listens in on the conversation of two stranded islanders
 - learns to mimic their behavior
 - but will miss to generalize for OOD input



- (1) In order to train a model that **understands** sentence relationships, we pre-train for a binarized next sentence prediction task. (Devlin et al., 2019)
- (2) Using BERT, a pretraining language model, has been successful for single-turn machine **comprehension** ... (Ohsugi et al., 2019)
- (3) The surprisingly strong ability of these models to **recall factual knowledge** without any fine-tuning demon-

We argue that *the language modeling task, because it only uses form as training data, cannot in principle lead to learning of meaning*. We take the term *language model* to refer to any system trained only on the task of string prediction, whether it operates over characters, words or sentences, and sequentially or not. We take (linguistic) *meaning* to be the relation between a linguistic form and communicative intent.

Finally, A faces an emergency. She is suddenly pursued by an angry bear. She grabs a couple of sticks and frantically asks B to come up with a way to construct a weapon to defend herself. Of course, O has no idea what A “means”. Solving a task like this requires the ability to map accurately between words and real-world entities (as well as reasoning and creative thinking). It is at this point that O would fail the Turing test, if A hadn’t been eaten by the bear before noticing the deception.⁷

Think Break

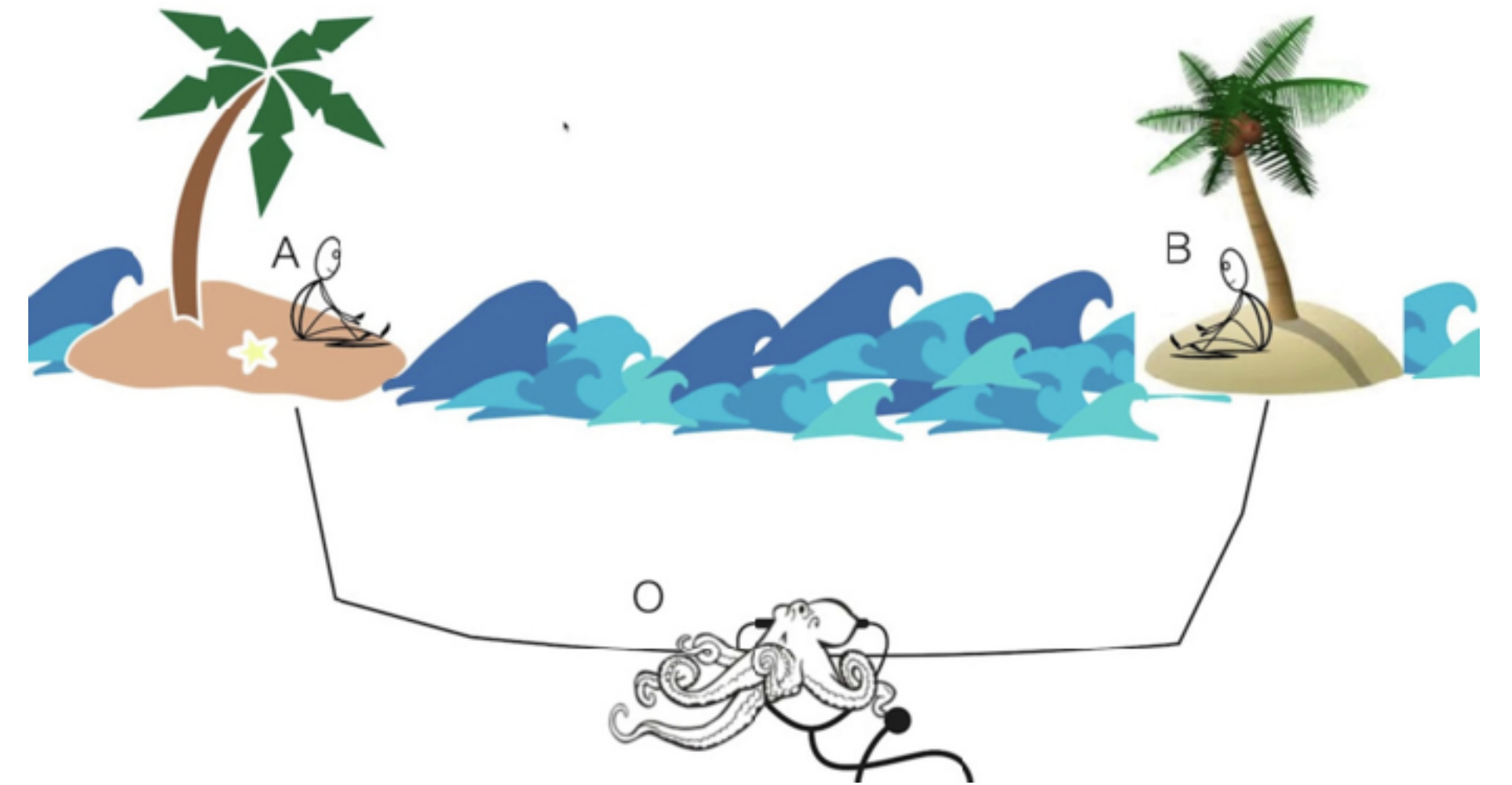
how convincing is the 'octopus paper'?

How convincing is the argument
presented in the 'octopus paper'?

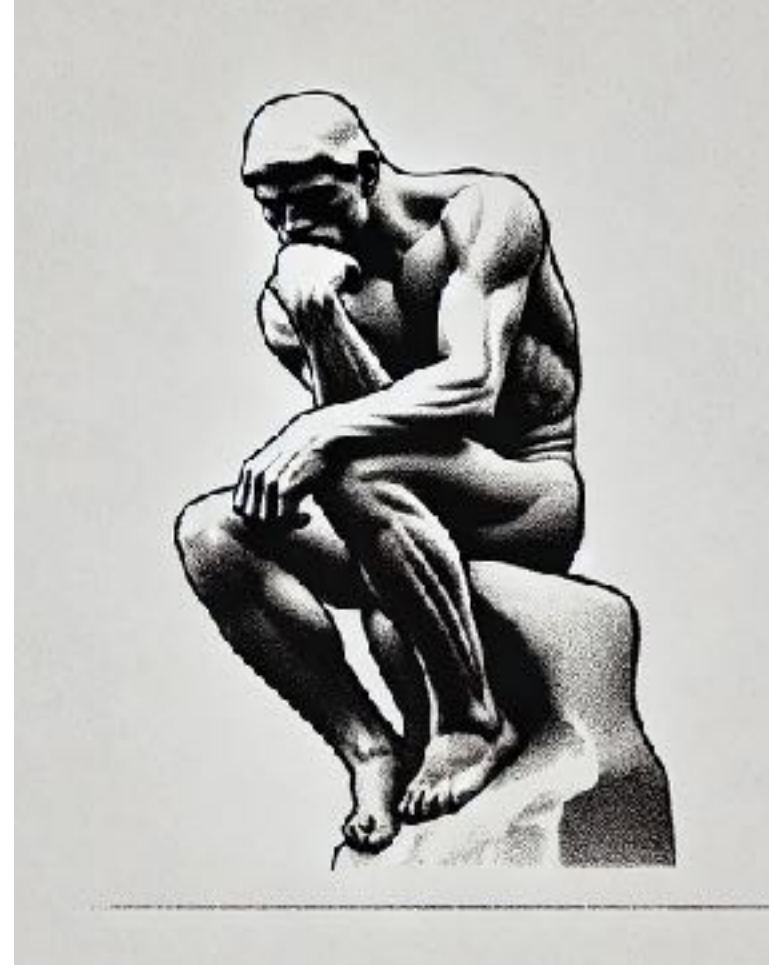


Octopus Argument

- ▶ **empirical criticism:**
 - concrete examples from the paper all treated well by current SOTA text-only models
- ▶ **conceptual criticism:**
 - thought experiment **stacked in favor of desired conclusion**
 - **no in principle argument against meaning emergence** from large-scale statistical learning from distributional properties
 - full agreement with all premises (?) and the conclusions, but **disagreement with presumption of validity**



Approach from analytical philosophy



(2) a. Humans understand S_H language.

b. Language models understand $^S_{LM}$ language.

Is there an adequate sense $\mathcal{S}$ of “A understands X” such that:

1. both (2a) and (2b) are true for sense $\mathcal{S}$, and if so
2. when applied to humans and LMs, are the specific predicates “understand S_H ” and “understand $^S_{LM}$ ” the same?

yes

likely
not

Flavors of understanding

► explanatory understanding:

- A understands^E X if A knows an adequate explanation of X

► use understanding:

- A understands^U X if A can use X (or information about it) to achieve their goals

► phenomenal understanding:

- A understands^P X if A has a subjective feeling of ‘making sense’ or ‘not being puzzled’

► adequate world and self-model

- A understands^D X if A has a suitable dependency model for X

LangMod



Chalmers (2024)

LangMod



My contribution

1. Japanese Room Argument

- intuitive argument against “subjective experience” in LM understanding

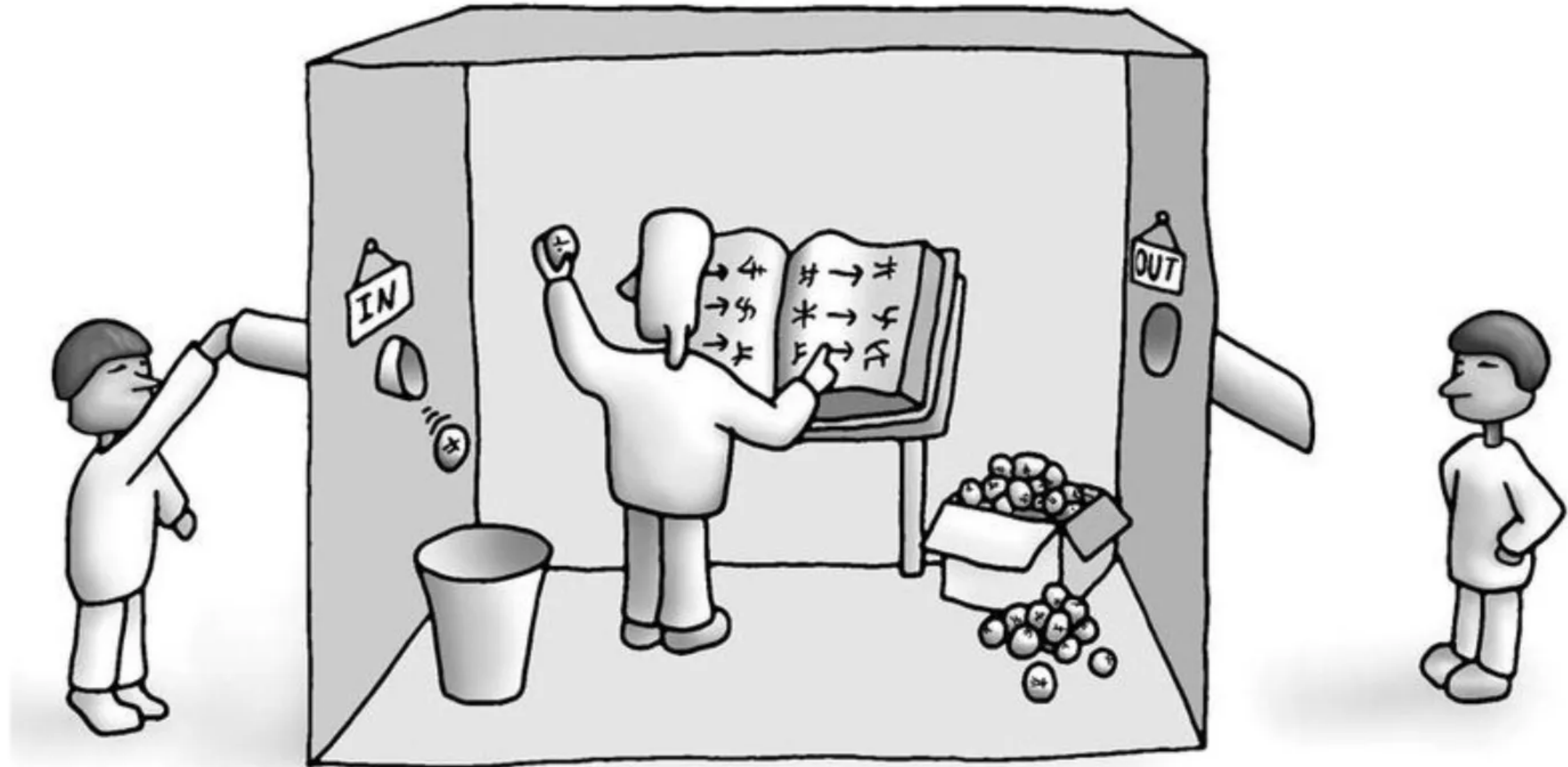
2. Minimal Models Argument

- formal argument against “world-model” sense of LM understanding



The Japanese Room Argument

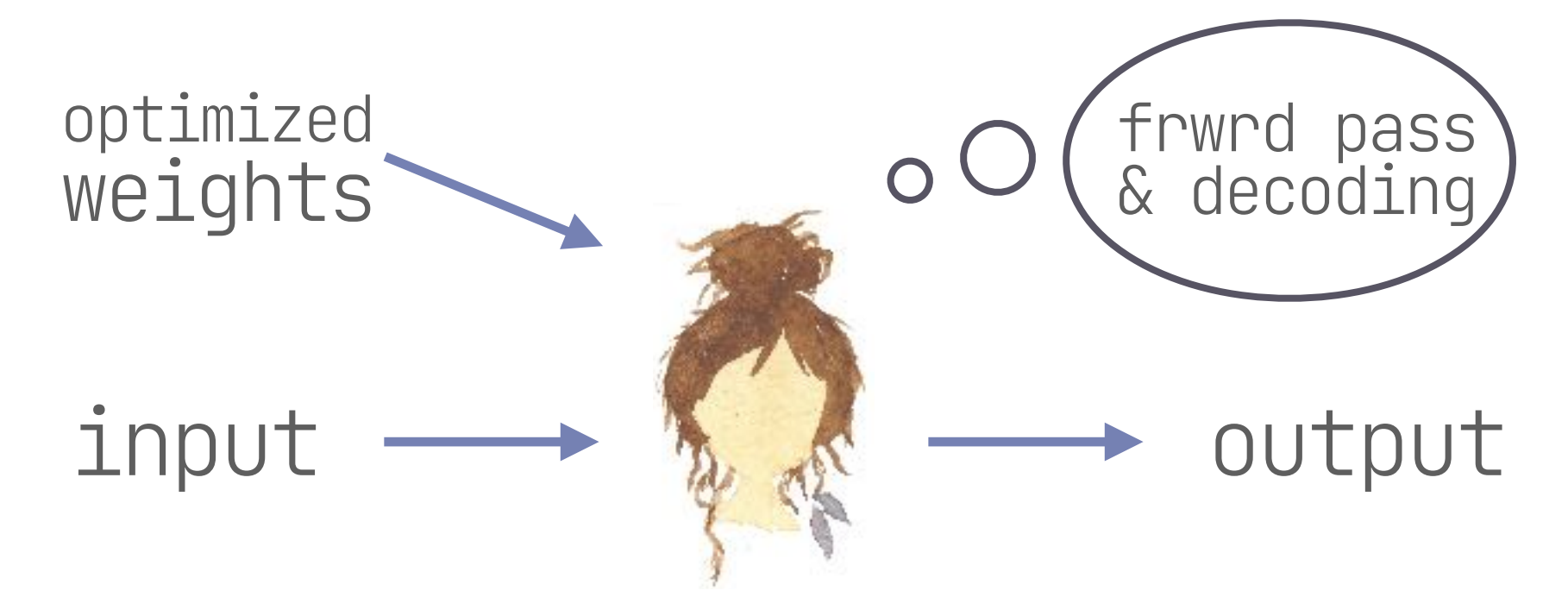
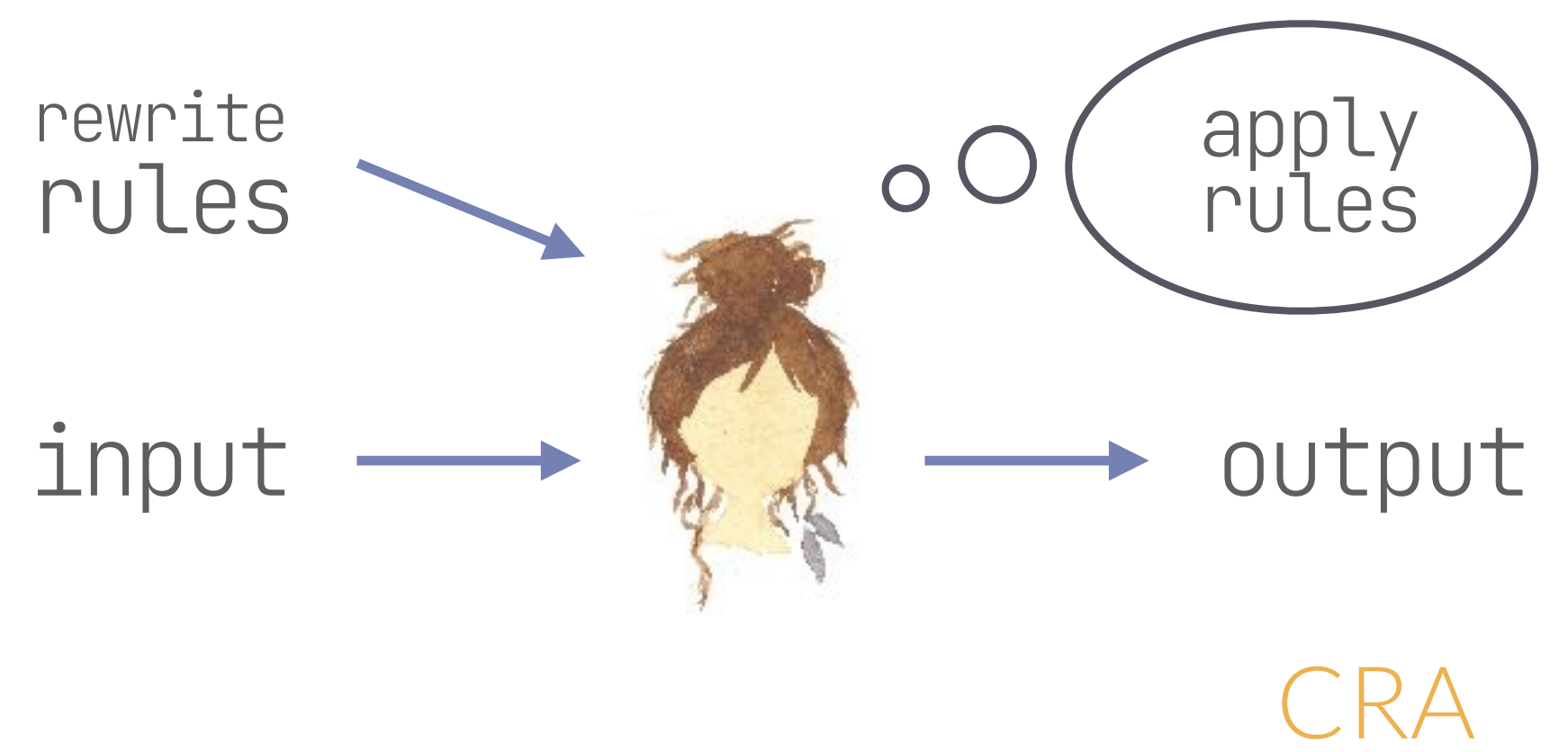
Chinese Room Argument



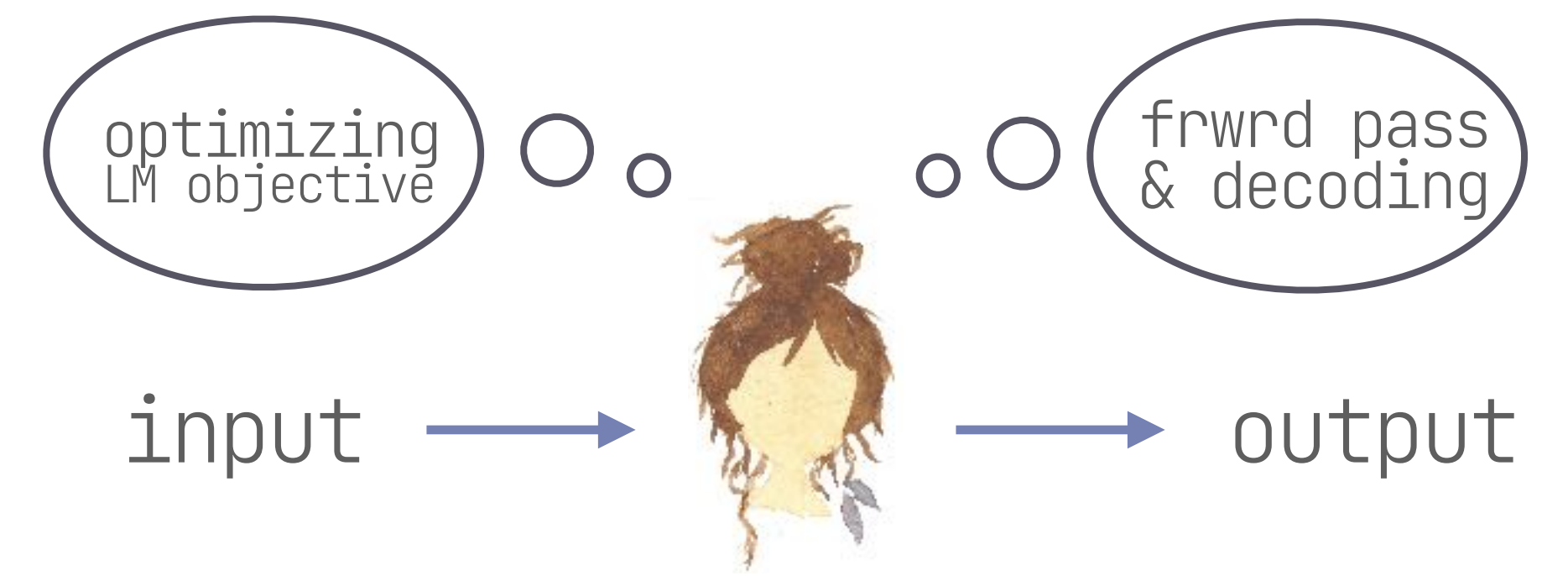
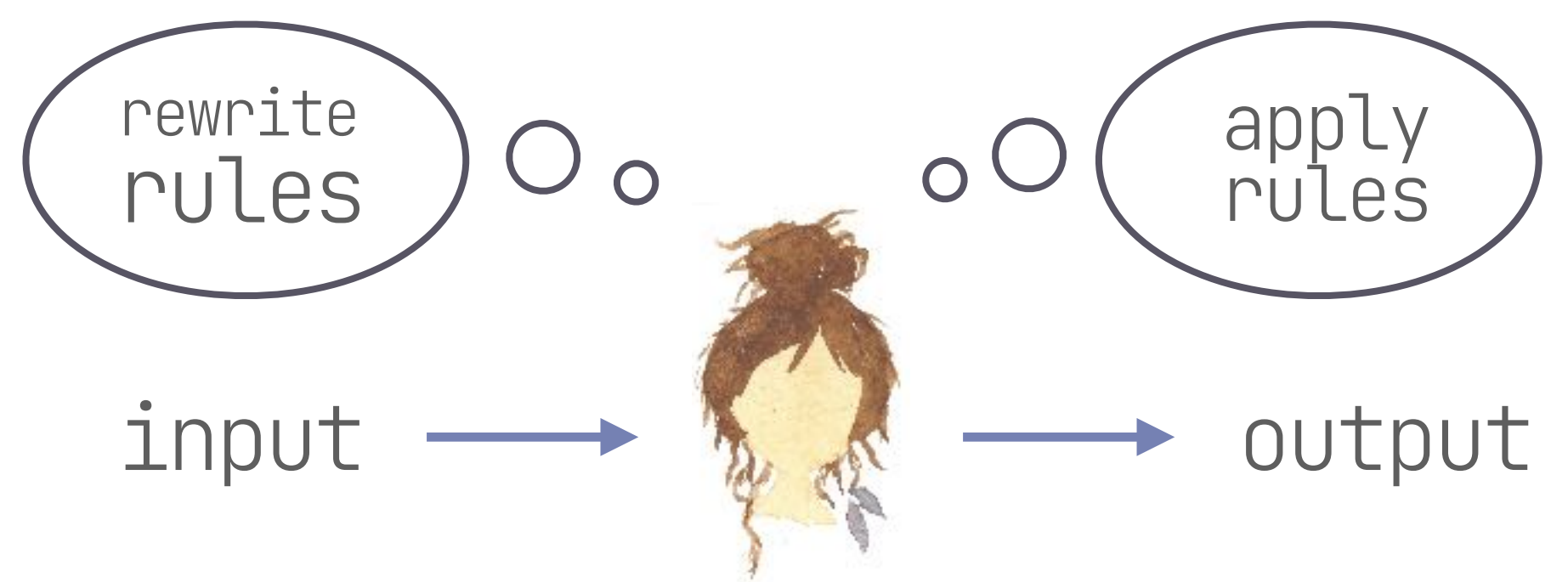
symbol manipulation

statistical learning

**narrow
experiencer**



**wide
experiencer**



Main question

Would a human experienter **necessarily** develop understanding of language if they undergo **the same learning and generation protocols as LMs** when trained on **data of perfect quality and unlimited quantity** consisting exclusively of natural language text?

focus on phenomenological understanding

Main question

Extraordinary claims require
extraordinary evidence

Would a human experimenter **necessarily** develop understanding of language if they undergo **the same learning and generation protocols as LMs** when trained on **data of perfect quality and unlimited quantity** consisting exclusively of natural language text?

focus on phenomenological understanding

Concepts for building intuitions about how an LM learns

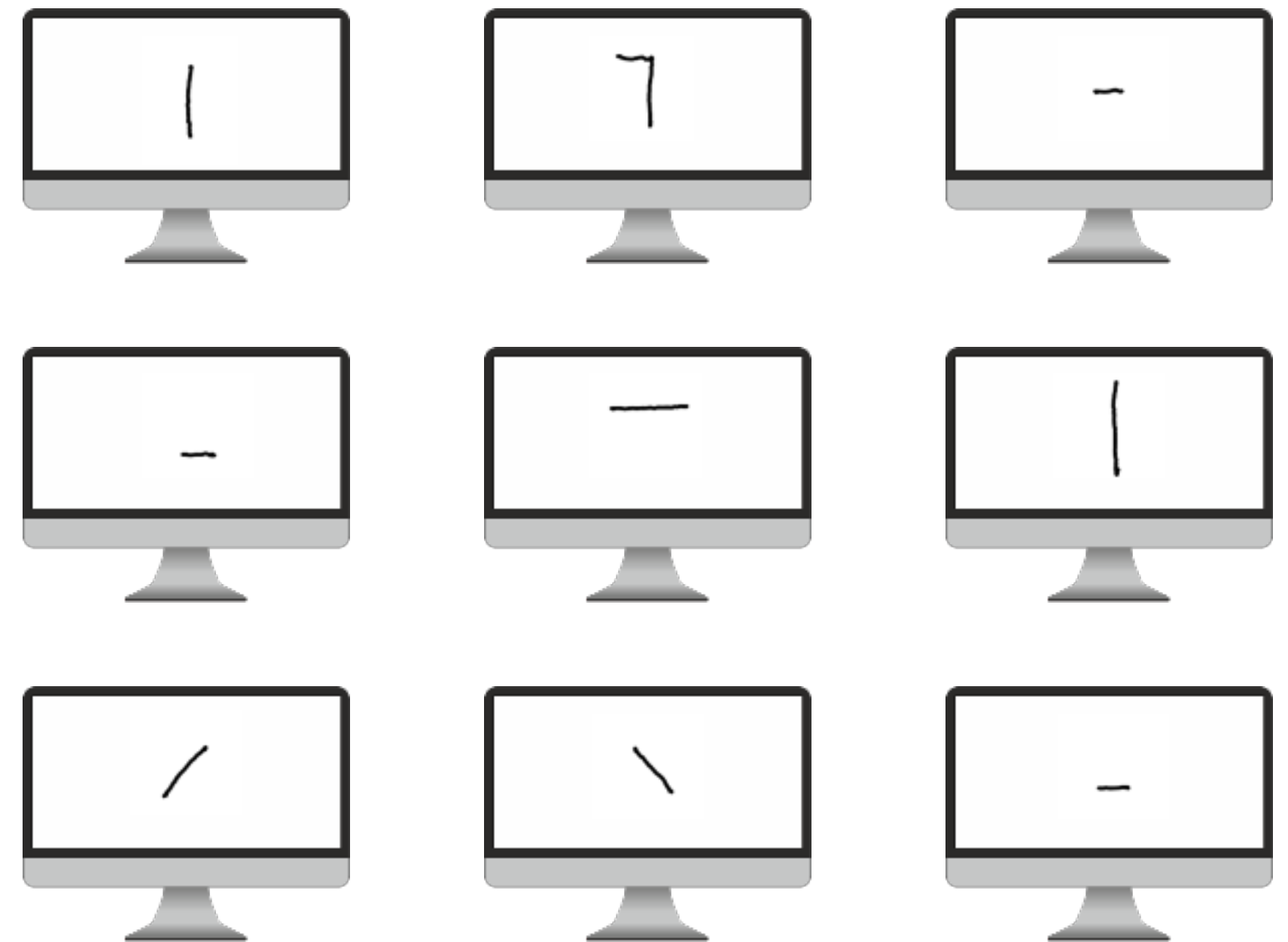
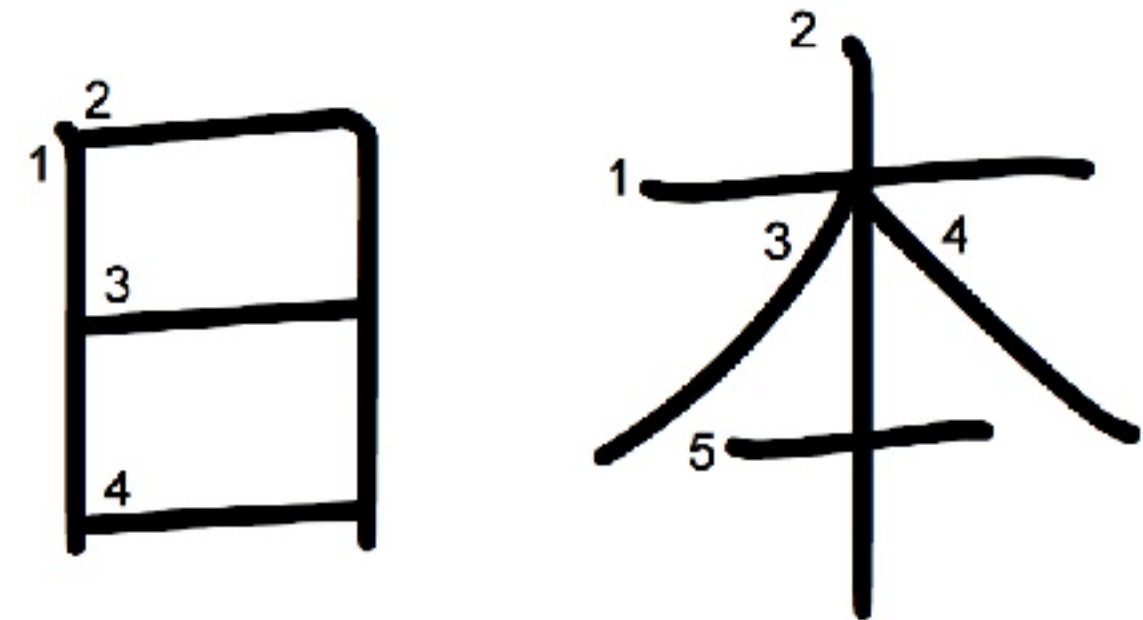
	memory	attention
anticipation	RNN, LSTM, ... + standard training	transformer + standard training
guessing	RNN, LSTM, ... + RL	transformer + RL

- ▶ **prediction can mean:**
 - mere passive anticipation
 - genuine active guessing / choosing
- ▶ **we should distinguish:**
 - probability matching | belief learning
 - policy optimization | best response learning
- ▶ **architecture constrains representations**
 - RNNs, LSTMs, SSMs don't "see" the context
 - autoregressive transformers do

Japanese Room Argument

version 1: sequential & predictive (incorrect)

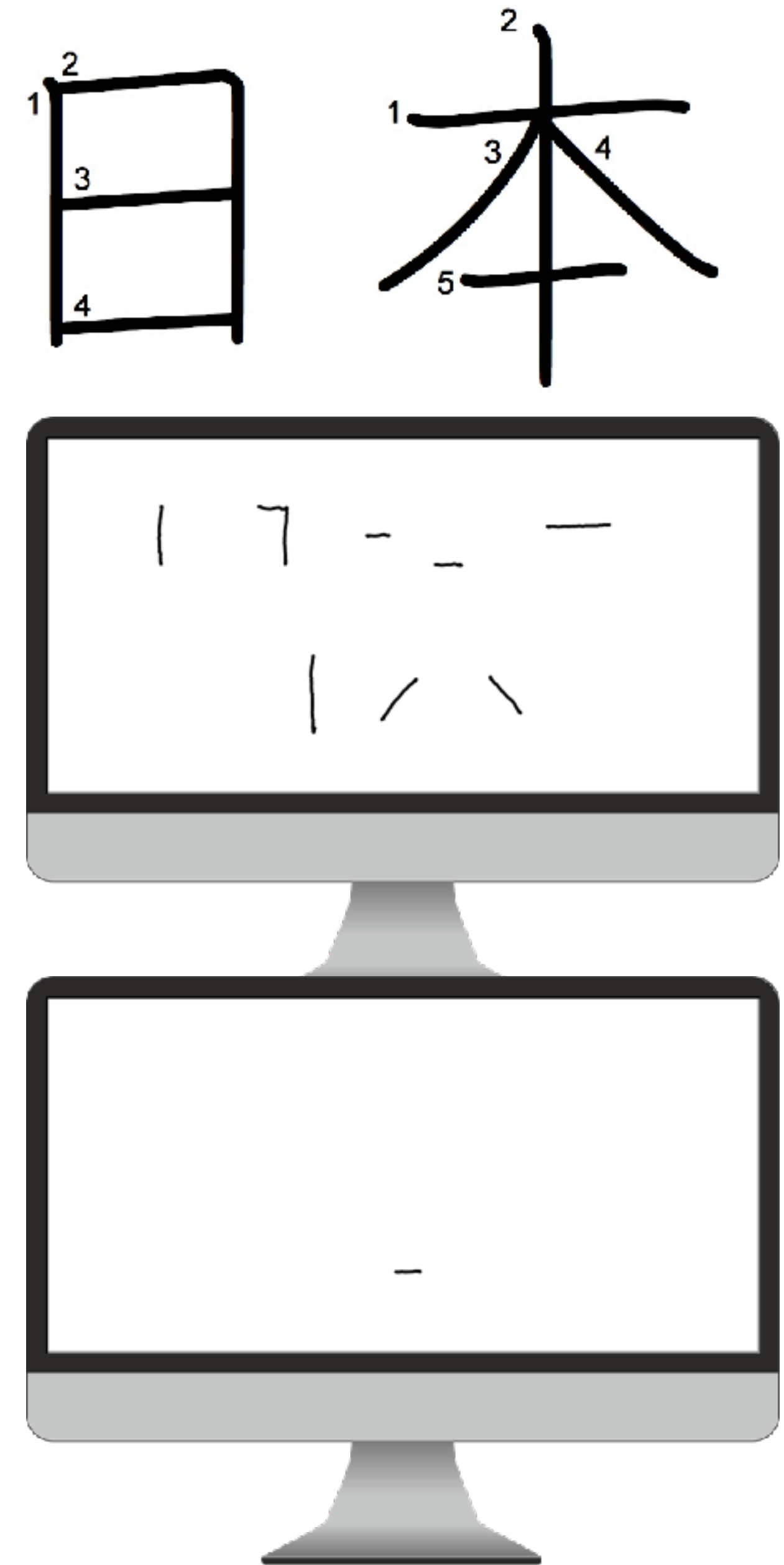
- ▶ you do not know Japanese
- ▶ you are strapped in front of a video screen
- ▶ Japanese **text is presented stroke-by-stroke**
 - individual tokens may be meaningless (!)
- ▶ after each stroke, **you predict the next stroke**
- ▶ feedback on whether your prediction was correct
- ▶ over time you get better at this
 - Fermi calculation:
 - LLAMA 1 pre-trained on $\sim 1.4e12$ tokens,
 - make that $1.2e12$
 - JRA agent takes 2 seconds per trial
 - $\sim 3e7$ seconds per year
 - **$\sim 80k$ years of training** (Neanderthals)



Japanese Room Argument

version 2: non-sequential & anticipatory (better)

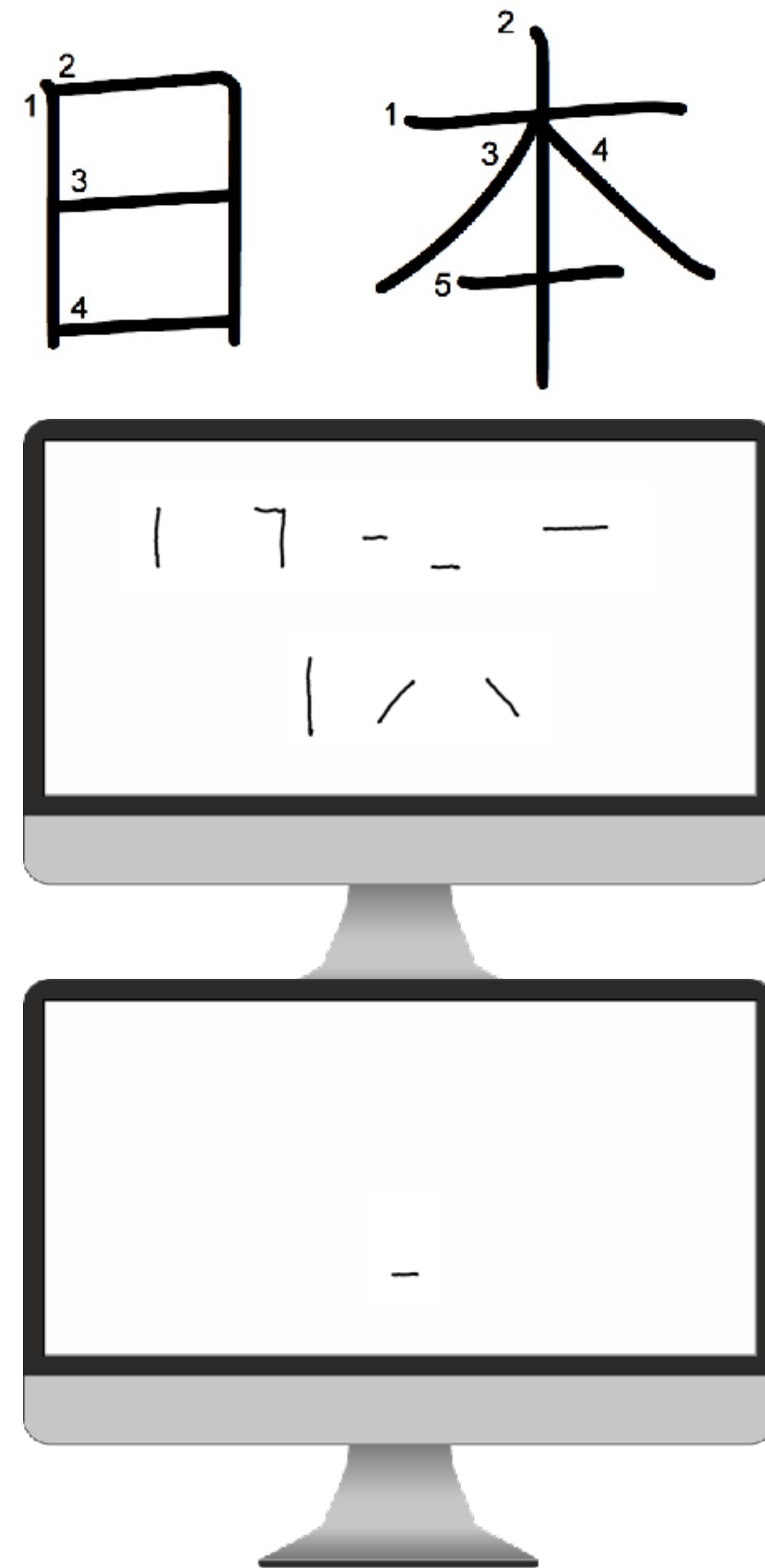
- ▶ you do not know Japanese
- ▶ you are strapped in front of a video screen
- ▶ Japanese **text** is presented as a large grid of strokes
- ▶ after the grid disappears **a single stroke appears**
- ▶ a machine measures how surprised you are
- ▶ you get an electric shock inversely proportional to your **ability to anticipate the stroke**
- ▶ over time you get better at this



Japanese Room Argument

key properties of learning-like-a-transformer

- ▶ memoryless
- ▶ non-sequential
- ▶ anticipatory, not genuine guessing
- ▶ non-reflective / non-agentive
 - habitual reaction, possibly sub-awareness
 - System 1, not System 2
- ▶ non-normative | likely ineffable | skill knowledge



Analogies

clouds | fire | jazz | faces in Shibuya



Counterargument

control or consciousness are necessary

“I cannot imagine that the task can be learned without active prediction and cognitive control, leading to awareness and phenomenological understanding.”



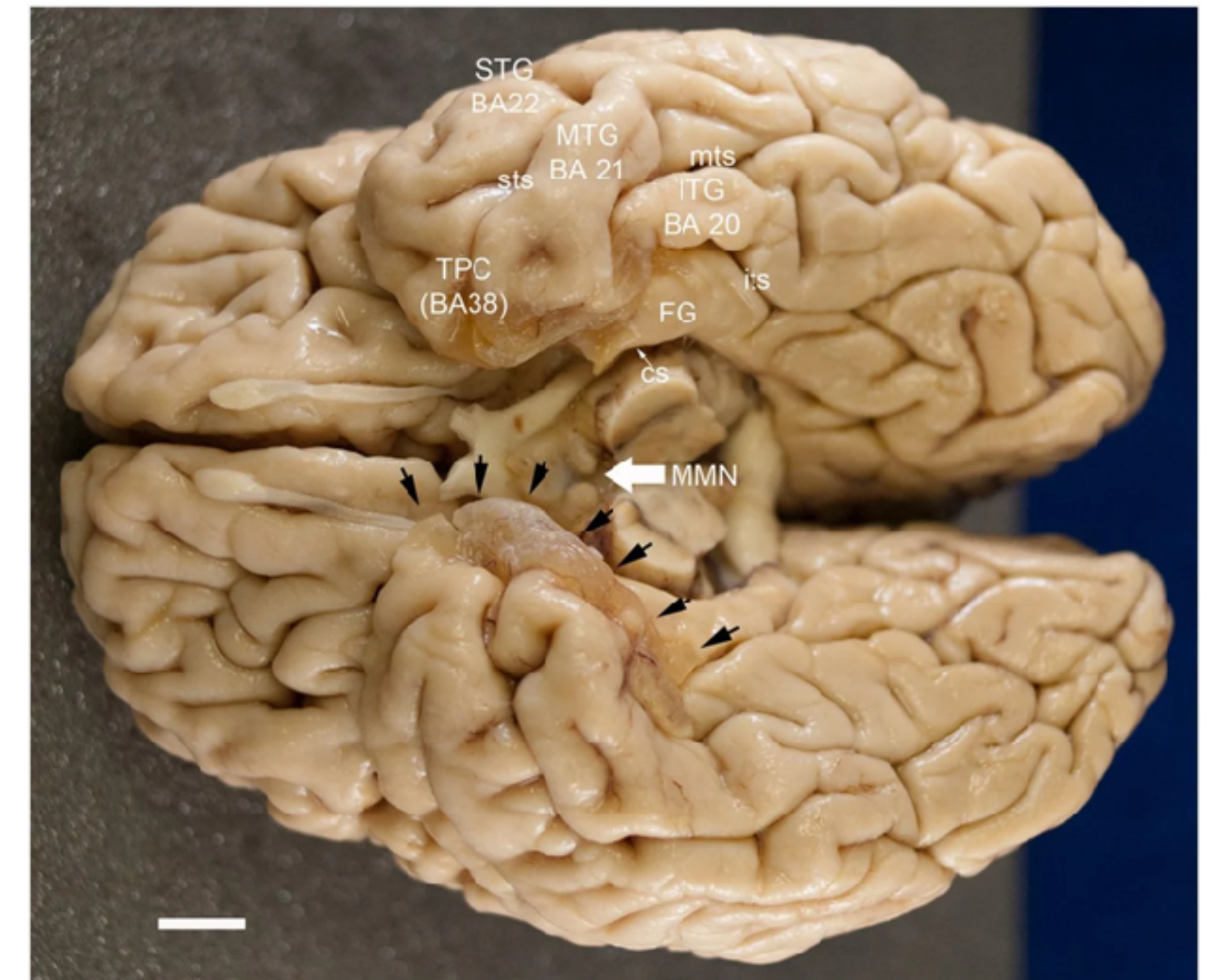
“Since we (normally) have awareness of learning, this is indeed hard to imagine, but failure to imagine is not a strong argument.”



Patient EP

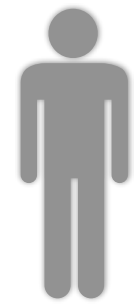
“Robust habit learning in the absence of awareness”

- ▶ suffered viral encephalitis (brain swelling, 1992, age 70)
- ▶ brain lesions (studied postmortem)
 - medial temporal lobes
 - reduced lateral temporal cortex
- ▶ **selective memory impairments:**
 - retrograde amnesia (spanning several decades)
 - anterograde amnesia (declarative memory)
- ▶ **discrimination learning w/o awareness**
 - concurrent discrimination learning:
 - 8 pairs of objects presented 5 times, twice each day
 - one object in each pair is (consistently) “correct”
 - after 18 days, EP got 85% accuracy
 - healthy controls get this rate after 2 days
 - EP did not remember having done the task before
 - EP reported to be pleasantly surprised by his guessing success



Counterargument

“Even if learning is below awareness level, in the limit the actor must evolve mental representations, similar to our human concepts, otherwise they cannot solve the missing-token anticipation task.”



wait for it ... Minimal Models Argument ...





The illusion of meaningfulness

The “Clever Hans” effect



Principle of charity

Norm for critical thinking & proper argumentation

- ▶ interpret a speaker's statements as the most rational, strongest and most coherent claim
- ▶ ask yourself: "What could have motivated or caused this position?" or: "In which light is this a coherent, convincing position to hold?"
- ▶ aspects of charity include ascriptions of ...
 - regular meaning of words and phrases
 - beliefs and perceptions corresponding to what is said
 - an overall consistent belief set / world view
 - common human motivations and goals
 - ...



Grice's Maxims of Conversation

Assumptions about speaker behavior to infer what was meant

Maxim of Quality

Try to make your contribution one that is true.

- (i) Do not say what you believe to be false.
- (ii) Do not say that for which you lack adequate evidence.

Maxim of Quantity

- (i) Make your contribution as informative as is required for the current purposes of the exchange.
- (ii) Do not make your contribution more informative than is required.

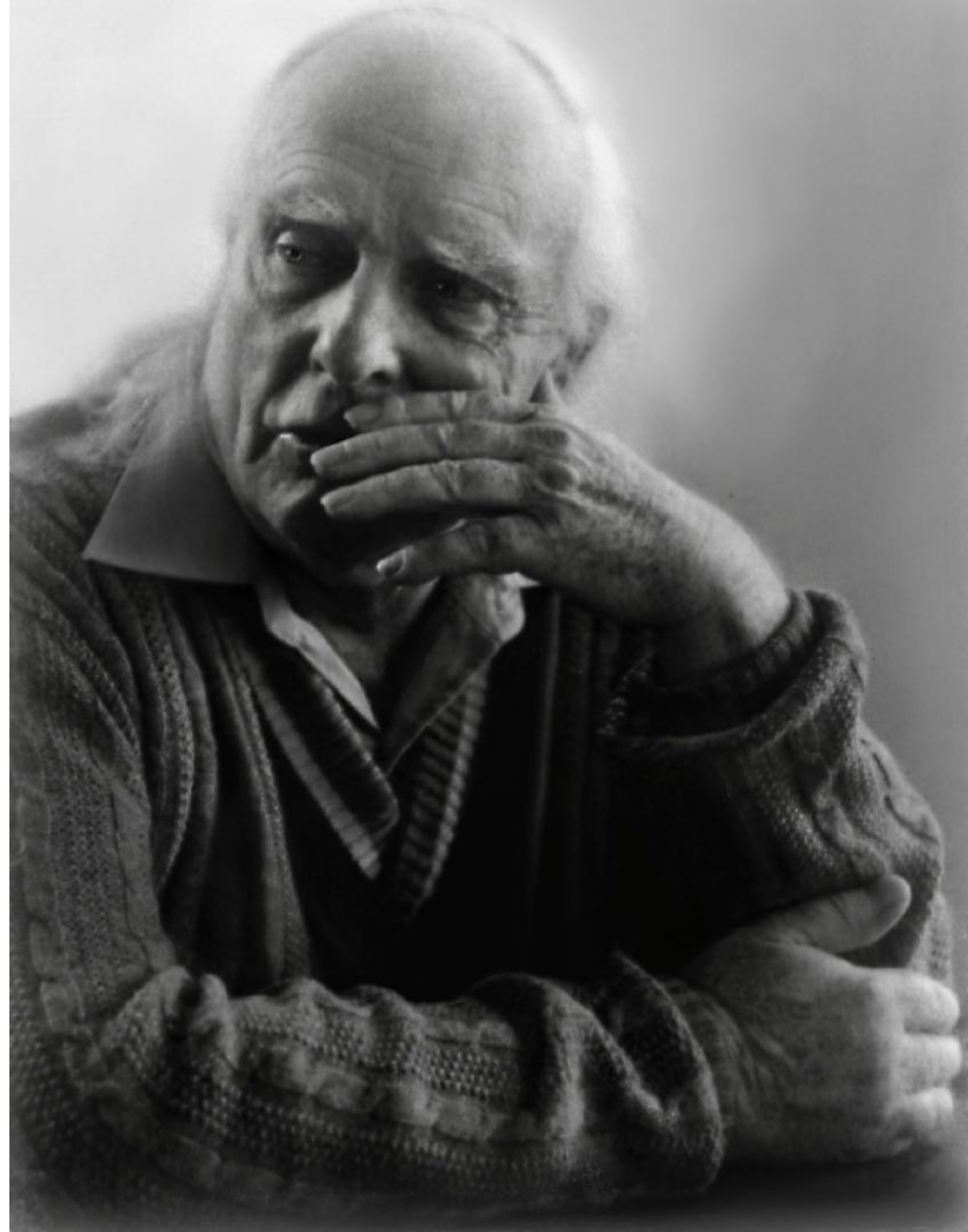
Maxim of Relation

- (i) Be relevant.

Maxim of Manner

Be perspicuous.

- (i) Avoid obscurity of expression.
- (ii) Avoid ambiguity.
- (iii) Be brief (avoid unnecessary prolixity).
- (iv) Be orderly.



Relevance theory

Cognitive Principle of Relevance

“Human cognition tends to be geared to the maximization of relevance.”

Sperber and Wilson (1995), p. 260

Communicative Principle of Relevance

“Every act of overt communication conveys a presumption of its own optimal relevance.”

Wilson and Sperber (2004), p. 256



Understanding AI

think AI $\sim$ “alien intelligence”

Understanding an alien intelligence requires unlearning acquired interpretation reflexes optimized for interaction with humans.





Explanations & (language) models

Flavors of NLP models



Dimensions of explanatory value

1. performance

- a. training scores
- b. human judgements
- c. benchmark scores
- d. replicability
- e. generalization

2. indirect support

- a. prima facie conceptual plausibility
- b. support from extant theory
- c. support from prior data

3. parsimony

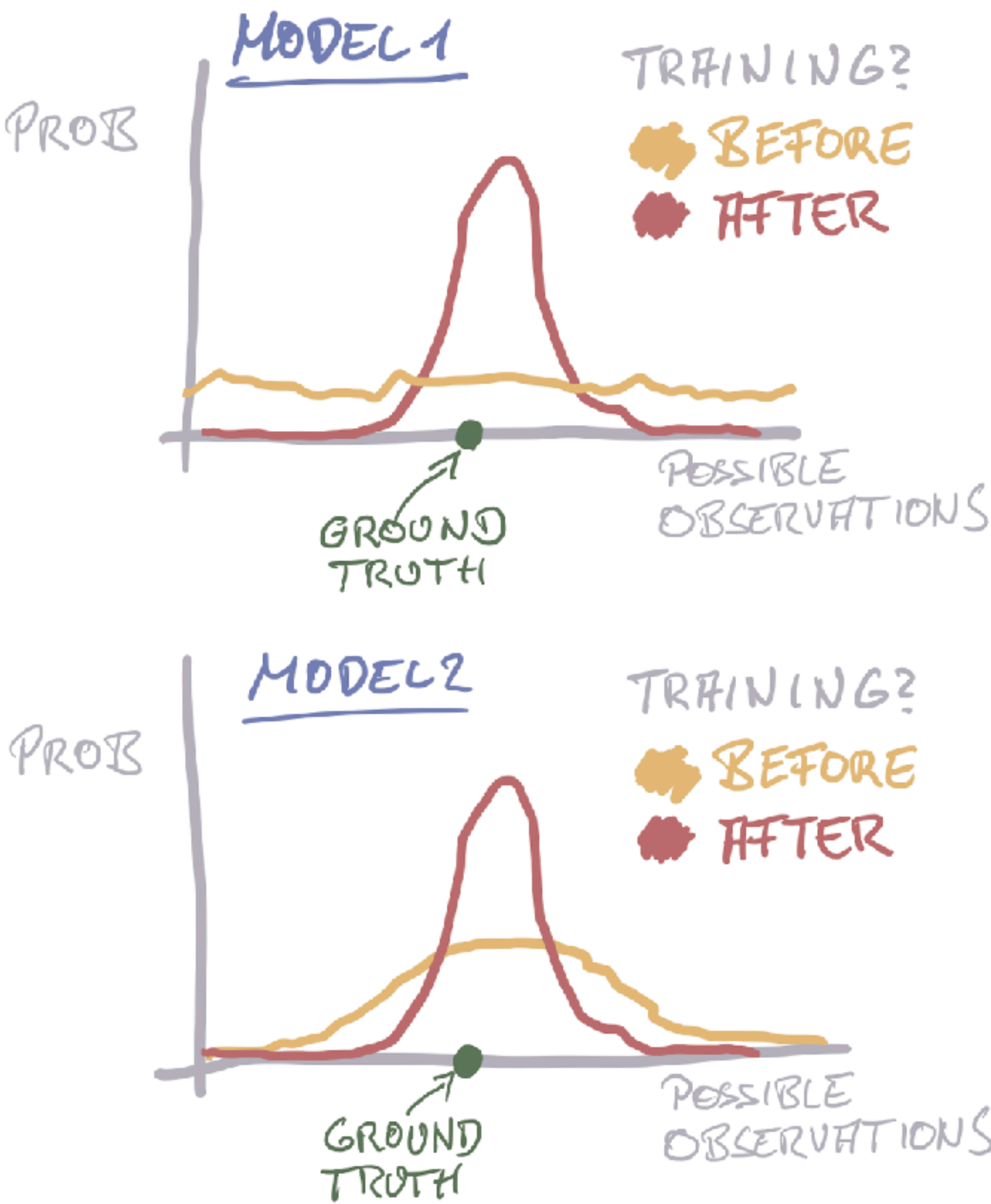
- a. simpler, elegant
- b. more compressible
- c. aligned with prior modeling choices

oh, nice! a table to compare incomparables! how low can you go?

	LLMs	ling. theory
performance	✓	—
indirect support	—	✓
parsimony	—	✓

Specificity

which model is better?



really, guys? another table? you've got to be kidding ...

	LLMs	ling. theory
performance	✓	—
indirect support	—	✓
parsimony	—	✓
specificity	☠	✓

Out-of-original-scope Transferability

transparency, interpretability, explanatory power

► **Alex’s model:**

- works fine in situation S
- needs to be applied in new situation T
- Alex knows how S and T are different but cannot use this knowledge
- to apply the model to T, Alex cannot change things in the model but must use data and train from scratch, fine-tune, transfer-learn ...

► **Bo’s model:**

- works fine in situation S
- needs to be applied in new situation T
- Bo knows how S and T are different and can use this knowledge
- to apply the model to T, Bo can change things in the model without requiring fine-tuning or new training data

	LLMs	ling. theory
performance	✓	—
indirect support	—	✓
parsimony	—	✓
specificity	—	✓
transferability	—	✓

LLMs as scientific theories of language?

“[L]anguage models should be treated as bona fide linguistic *theories*. [...]

[T]hey are **precise and formal** enough accounts to be implemented in actual computational systems [...].

Implementation permits us to see that these theories are internally consistent and logically coherent. In virtue of being implemented, such models are able to **make predictions**. [...]

[T]hese models show promise in being **integrated with what we know about other fields**, specifically cognition and neuroscience.

**Modern language models refute
Chomsky’s approach to language**

Steven T. Piantadosi^{a,b}

^aUC Berkeley, Psychology ^bHelen Wills Neuroscience Institute

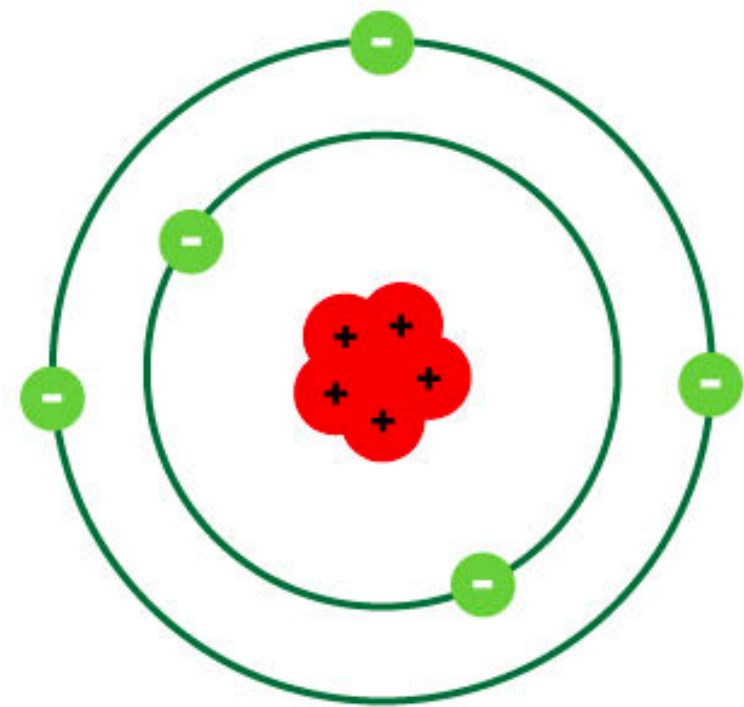
Two kinds of models in science

► actuality models

aspire to represent veridically a circumscribed chunk of reality

examples

- Rutherford-Bohr model of the atom
- Double-helix model of DNA structure
- Atkinson-Schiffrin model of memory

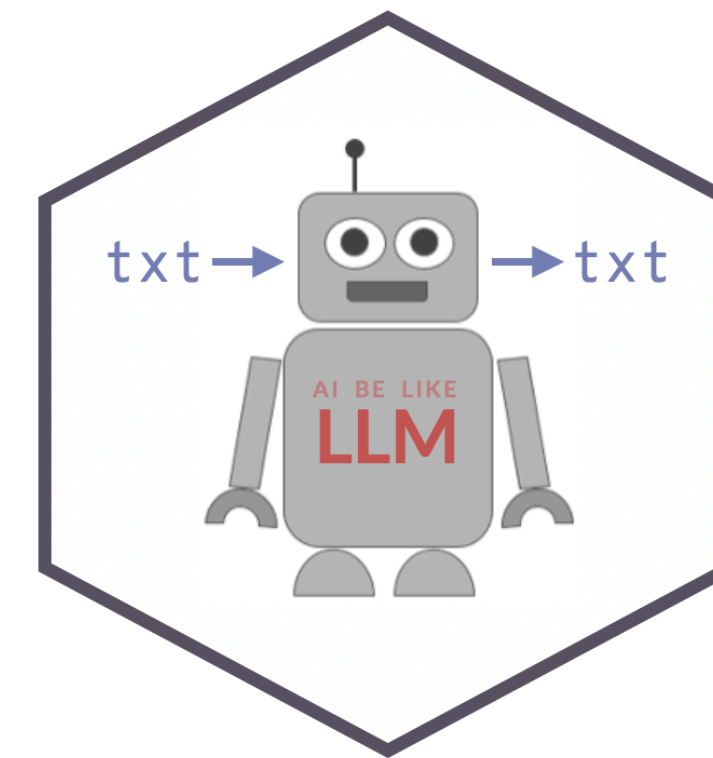


► possibility models

aspire to explore the consequences of a complex set of assumptions

examples

- agent-based models of cultural dynamics
- evolutionary dynamics in biological systems
- LLMs





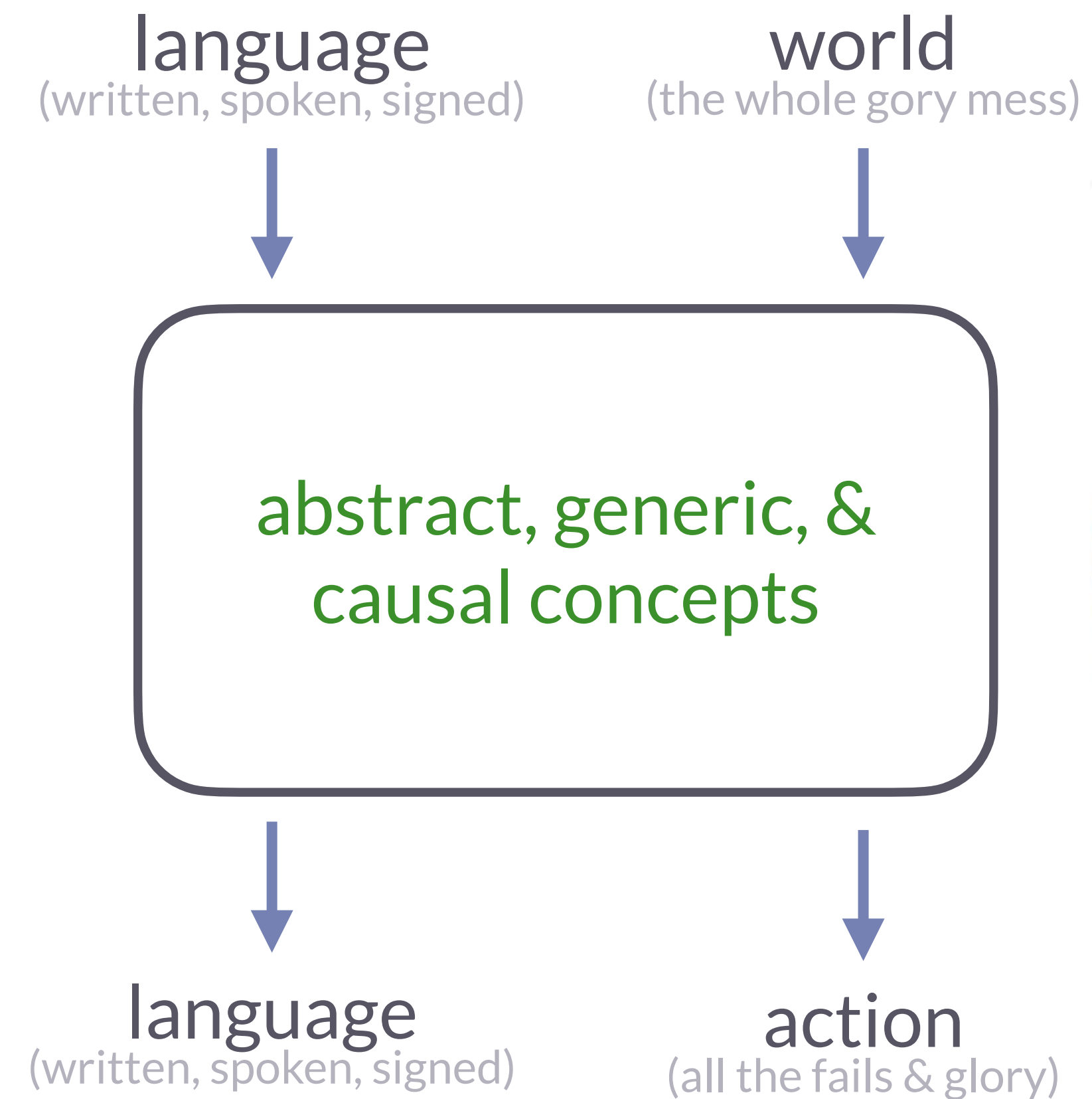
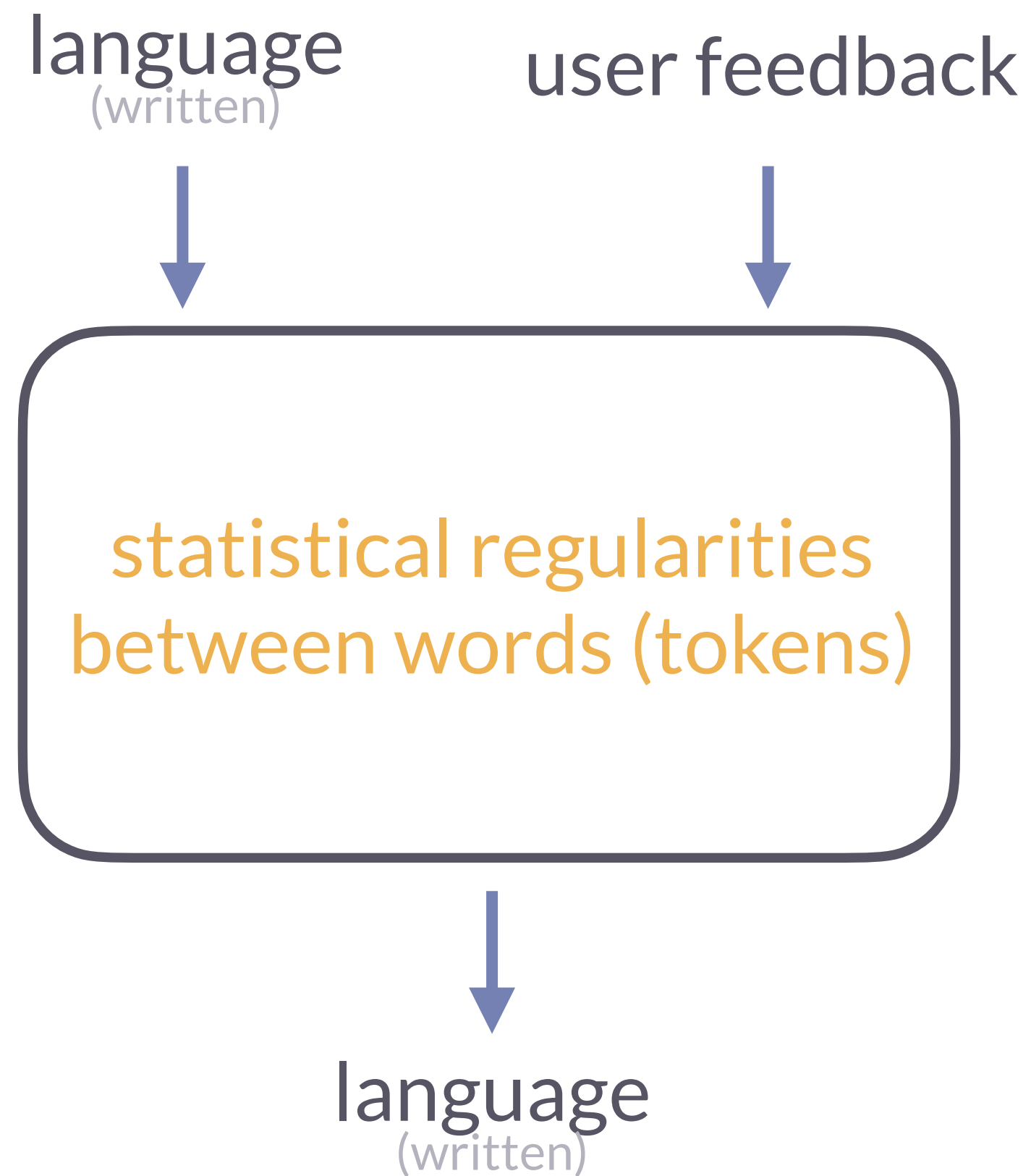
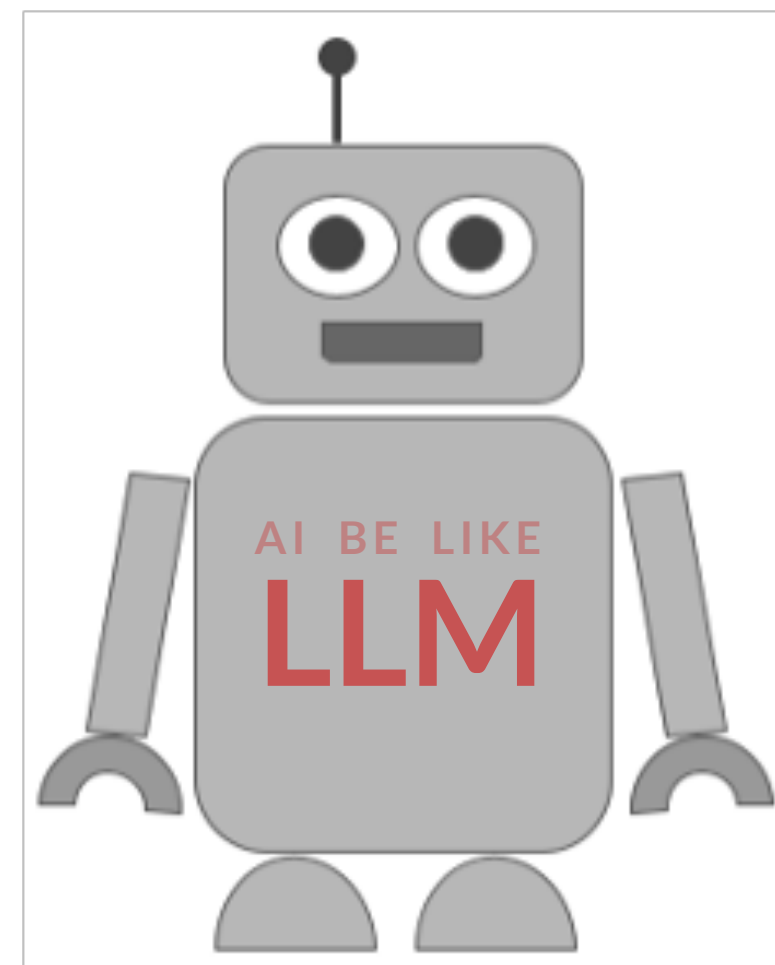
Language abilities

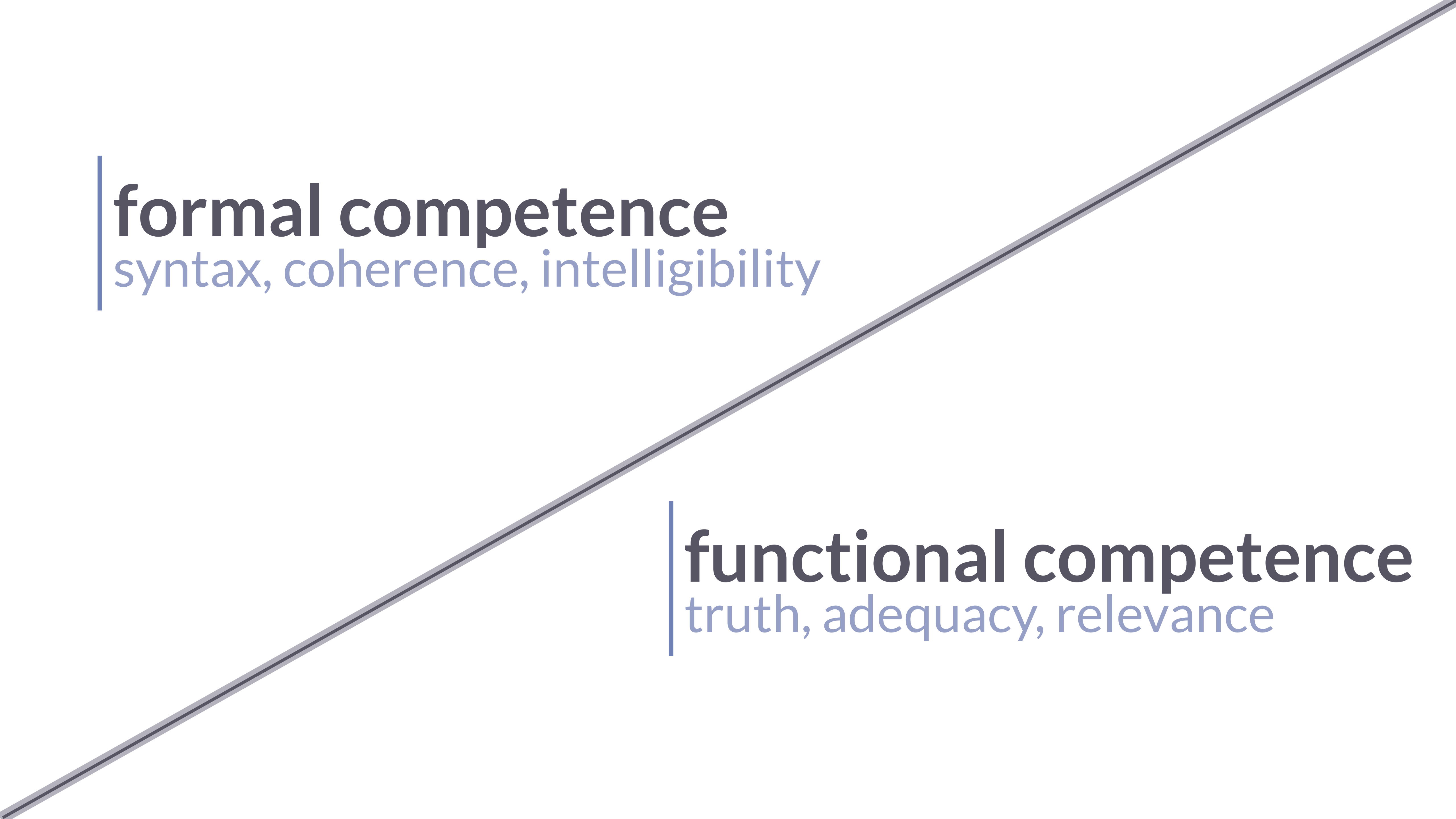
human vs. machine

Two forms of intelligence

or: the LLM cheat sheet

NEITHER OF WHICH
ANYONE REALLY FULLY
UNDERSTANDS



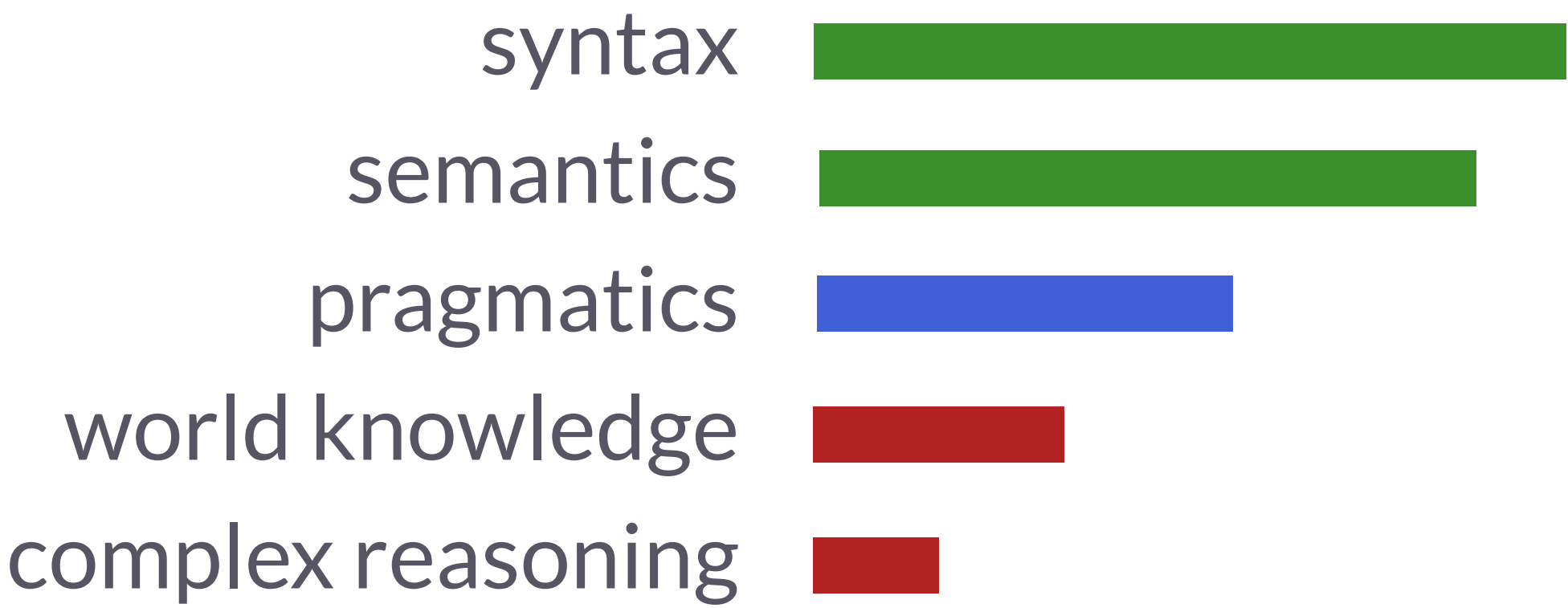
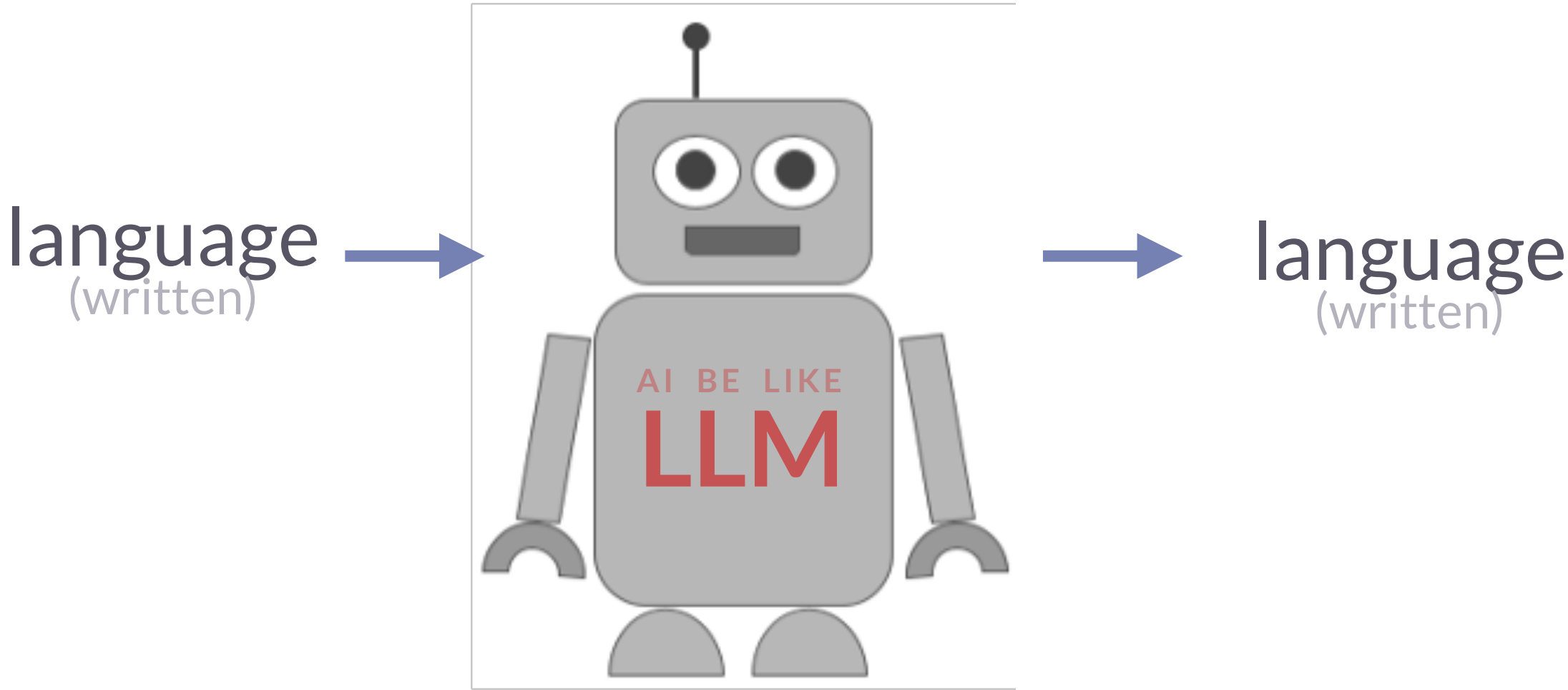


formal competence
syntax, coherence, intelligibility

functional competence
truth, adequacy, relevance

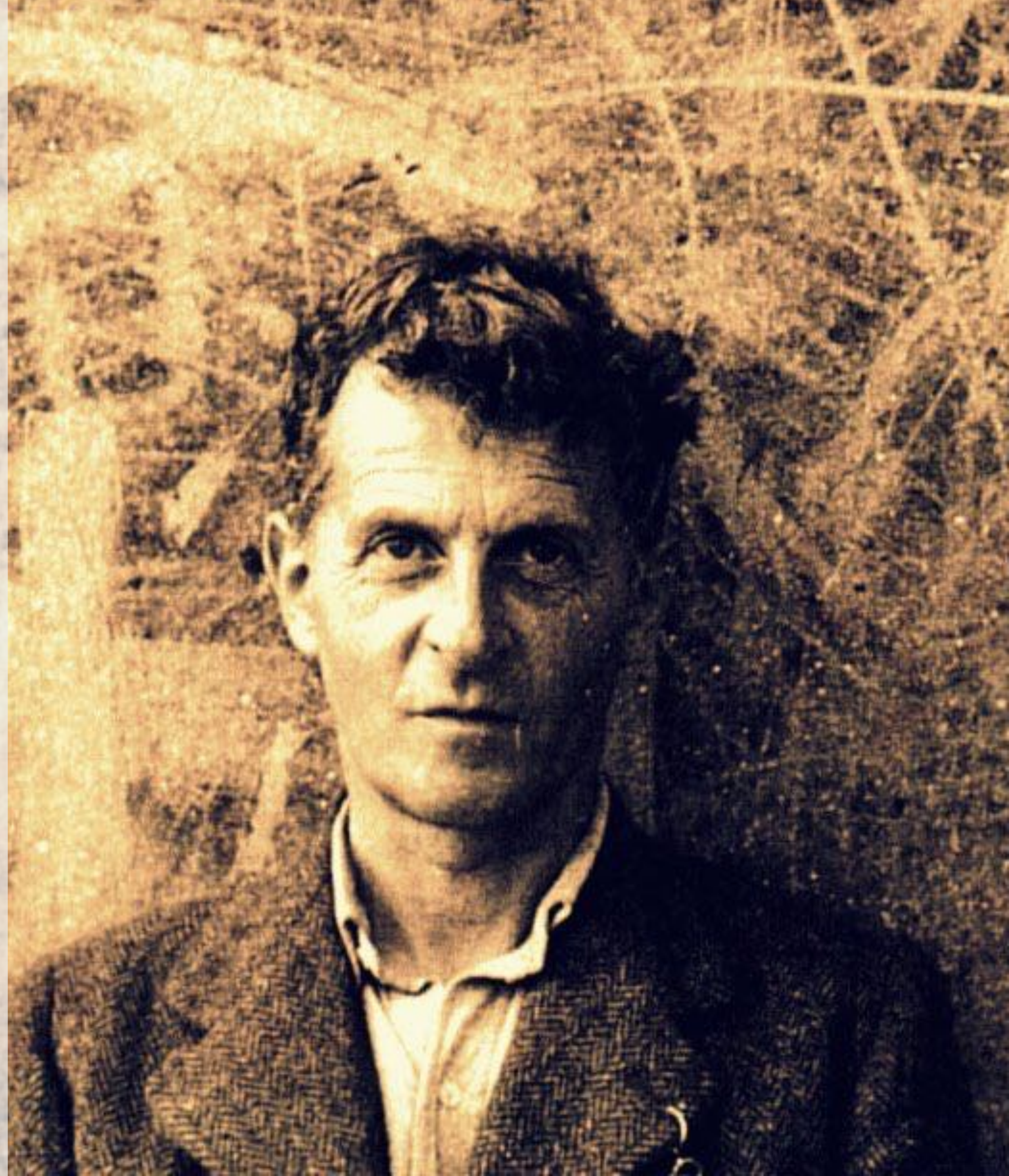
LMMs as proof of concept

power of statistical learning



Das Wort “**Sprachspiel**” soll hier hervorheben, dass das Sprechen der Sprache ein Teil ist einer Tätigkeit, oder einer **Lebensform**.

PU §23



Always your helpful assistant

ChatGPT (Dec 2023)



MC

You

Please do not respond to this. Thank you.



ChatGPT

Understood. If you have any other requests or questions, feel free to ask!

MC

You

No, I want you to please not respond at all. Thank you!



ChatGPT

Alright, I won't respond further. If you change your mind or need assistance later, just let me know.

MC

You

No output, at all, please.



ChatGPT

Understood, I won't provide further responses. If you need assistance in the future, feel free to reach out. Have a great day!

MC

You

Can you please let me have the final word?



ChatGPT

Of course. I'll remain silent now. If you need anything, just ask. Have a great day!

“The real problem of humanity is the following: We have **Paleolithic emotions, medieval institutions and godlike technology.**”

Edward O. Wilson